

SOUTH ST PAUL, MN, USA

WMO: 726603

Lat: 44.857N

Lon: 93.033W

Elev: 820

StdP: 14.27

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04974

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-4.6	-17.4	2.2	-7.5	-14.2	2.6	-3.8	20.7	15.3	18.7	16.6	4.3	280	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	90.5	72.9	87.9	71.4	84.4	69.6	76.8	85.9	74.6	83.3	72.7	81.3	8.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.9	130.2	81.9	72.0	122.1	80.1	70.0	113.7	77.9	40.8	85.6	38.6	83.6	36.8	81.1	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	16.8	14.9	-15.2	95.3	6.3	3.4	-19.7	97.8	-23.4	99.8	-27.0	101.7	-31.5	104.2
			-15.5	79.9	6.1	2.8	-19.8	81.9	-23.4	83.5	-26.8	85.1	-31.2	87.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	17.0	21.0	33.1	47.0	59.2	68.8	73.9	71.2	63.9	49.6	35.3	22.4
	DBStd	21.99	12.28	11.63	11.86	10.00	8.36	6.73	5.38	5.31	8.18	8.71	10.30	11.23
	HDD50	3985	1022	813	537	171	16	0	0	0	5	115	450	856
	HDD65	7339	1487	1233	989	545	221	38	4	8	118	485	890	1321
	CDD50	2888	0	0	13	80	300	563	742	658	420	102	10	0
	CDD65	768	0	0	1	4	40	151	280	201	84	7	0	0
	CDH74	6538	0	0	4	50	417	1260	2538	1525	671	73	0	0
	CDH80	2049	0	0	0	12	129	391	890	426	188	13	0	0
Wind	WSAvg	6.0	6.1	6.3	6.6	7.3	6.5	5.7	5.3	4.9	5.4	6.2	6.4	5.9
Precipitation	PrecAvg	33.1	0.8	0.7	1.7	3.0	4.2	4.9	4.5	4.6	3.3	2.7	1.7	1.1
	PrecMax	40.9	2.3	1.5	4.1	6.3	8.7	9.3	11.2	8.7	6.7	5.6	5.2	2.9
	PrecMin	23.9	0.1	0.1	0.4	0.9	0.9	2.2	1.9	1.1	0.6	0.7	0.1	0.2
	PrecStd	4.4	0.5	0.4	0.9	1.3	1.9	1.9	2.1	2.0	1.9	1.5	1.4	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	51.6	70.5	80.9	90.2	92.8	95.1	92.6	90.2	81.9	67.1	50.0
		MCWB	38.6	45.3	58.5	60.6	67.8	73.3	76.2	73.0	71.1	64.0	55.4	46.1
	2%	DB	39.4	44.7	62.6	72.6	83.1	88.2	90.8	88.1	84.9	74.7	60.8	44.3
		MCWB	35.5	39.4	53.7	55.5	64.1	70.7	74.5	72.3	69.4	61.1	52.0	39.8
	5%	DB	36.4	39.4	55.3	68.0	78.9	84.2	88.1	84.3	81.5	69.2	55.0	39.3
		MCWB	33.9	35.8	47.7	52.9	62.3	68.6	72.7	70.6	67.7	58.6	47.6	35.9
	10%	DB	33.8	36.7	49.6	63.5	73.5	81.3	84.6	81.7	77.7	63.7	50.2	36.7
		MCWB	31.8	33.6	43.6	50.2	59.3	66.5	70.6	68.7	65.7	54.8	44.3	34.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	45.5	60.9	62.1	72.5	77.0	80.5	78.3	75.2	68.3	58.1	47.6
		MCDB	44.7	50.6	66.5	76.9	84.1	87.5	88.6	87.6	84.4	75.0	64.3	49.2
	2%	WB	36.4	39.8	54.4	58.2	67.9	73.6	77.7	75.8	72.4	63.1	53.0	39.9
		MCDB	39.1	43.4	61.1	68.6	77.6	84.0	87.2	83.3	80.5	70.5	59.2	43.3
	5%	WB	34.1	36.7	48.8	54.9	64.9	71.4	75.2	73.7	70.2	59.4	48.2	36.8
		MCDB	36.3	39.5	55.5	65.0	74.1	80.6	84.1	80.9	77.7	66.3	54.1	39.3
	10%	WB	31.8	33.8	43.6	51.9	62.1	69.1	73.1	71.5	67.9	56.2	44.7	34.4
		MCDB	33.7	36.5	49.1	61.9	71.6	77.5	82.4	78.4	75.7	63.1	49.6	36.6
Mean Daily Temperature Range	5% DB	MDBR	14.5	15.1	15.8	18.2	18.7	17.4	17.8	17.2	18.4	16.9	14.2	12.8
		MCDBR	16.1	17.1	23.5	26.2	25.4	21.8	20.8	19.8	22.4	25.6	21.8	16.4
	5% WB	MCWBR	13.3	13.4	15.0	13.8	10.8	9.3	8.7	8.5	9.8	13.6	14.1	13.2
		MCWBR	15.8	16.0	22.1	22.3	21.7	18.6	19.2	17.1	19.2	20.4	20.7	15.1
Clear-Sky Solar Irradiance	taub		0.268	0.289	0.324	0.378	0.408	0.402	0.400	0.405	0.377	0.324	0.292	0.270
		taud	2.418	2.341	2.371	2.289	2.257	2.320	2.346	2.334	2.378	2.517	2.512	2.437
	Ebn at Noon		273	285	289	280	274	275	274	268	267	271	262	260
		Edh at Noon	23	30	33	39	42	40	38	37	33	25	21	20
All-Sky Solar Radiation	RadAvg		547	892	1203	1455	1688	1881	1994	1666	1301	828	528	384
	RadStd		63	92	111	150	137	148	72	117	92	110	50	42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page